

SACHIN SONI

☎ +91 6386732065

✉ sachinsoni600517@gmail.com

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Technical Skills

Languages and Databases: Python, MySQL, PostgreSQL, FAISS, Pinecone

Frameworks: PyTorch, FastAPI, LangChain, LangGraph

AI/ML Domains: Machine Learning, Deep Learning, GenAI, Agentic AI, RAG, MLOps

Tools and MLOps: Git, Docker, GitHub Actions, GitLab CI/CD, MLFlow

Cloud: AWS

Experience

Senior AI Engineer

Tata Elxsi

November 2025 – Current

Bengaluru, Karnataka

Sketch to 3D Model Generator

- Built an AI pipeline that converts 2D industrial sketches (e.g., coolers) into high-quality images using Stable Diffusion, applying text-based conditioning prompts to control shape, material, and design details.
- Integrated these generated images with a 3D model generator (e.g., **Trellis**) to create 3D models, and reducing manual CAD modeling efforts to our business unit.

AI Engineer

Tata Elxsi

July 2023 – October 2025

Bengaluru, Karnataka

RAG ChatBot

- Developed a production-grade multimodal RAG chatbot capable of processing PDFs with text, tables, and images using Docling, deployed on AWS EC2 with **PostgreSQL + pgvector** for fast vector search.
- Cut LLM inference costs by **60%** through **semantic caching**, token limit, and intelligent rate limiting—while maintaining consistent response quality.

Vision Inspection

- Built an AI-driven vision inspection system using **YOLOv7** to automatically detect defective medical-kit packets on a conveyor line for a Japanese client, enabling automated quality control.
- Addressed a **400:200 class imbalance** using Albumentations-based targeted augmentation, improving recall for defective samples and reducing false negatives.
- Fine-tuned YOLOv7 with COCO-pretrained weights, achieving **91% mAP@0.5** and **25–30 FPS** real-time inference on GPU.

AI Containerization

- Standardized development environments using DevContainers and implemented a high-performance **gRPC** client-server framework with expanded unit/integration tests to improve reliability.
- Integrated **mypy**, **ruff**, **bandit**, and automated security scans into GitLab CI/CD; engineered multi-stage Docker builds that reduced image size by **60%** and pipeline time by **35%**.

AI Engineer Intern

Tata Elxsi

Jan 2023 – June 2023

Bengaluru, Karnataka

Food Delivery Time Prediction

- Built an end-to-end regression pipeline to predict food delivery time (in minutes) using 1 Lakh+ real-world orders with features like weather, traffic, and distance etc.
- Engineered time-based and geospatial features (Haversine distance, time-of-day, festivals) and applied **Bayesian Optimization** to fine-tune XGBoost Random Forest models, **reducing MAE from 5 minutes to 3 minutes**.
- Developed a **Stacking Regressor** combining XGBoost and Random Forest with Linear Regression meta-learner for best accuracy.

Education

National Institute of Technology,Patna <i>Master of Computer Applications (MCA)</i>	Oct 2021 – June 2023 <i>Patna, Bihar</i>
Harish Chandra Post Graduate College <i>BSc Maths</i>	July 2018 – June 2021 <i>Varanasi, Uttar Pradesh</i>

Certifications

- Oracle Cloud Infrastructure Generative AI Professional – Oracle (2025)

Medium Technical Blog Writer

- Authored **80+** in-depth blogs on topics including RAG, MCP, Transformers, ML, DL, Docker, and MLOps.
- Gained **800+**followers and **9K+** monthly reads, helping simplify complex concepts for a global audience.

Achievements

- Secured an All India Rank (AIR) of **3614** in **GATE 2024** (Data Science and Artificial Intelligence) and All India Rank (AIR) of **675** in **NIMCET 2021** (National Institute of Technology MCA Entrance Test).